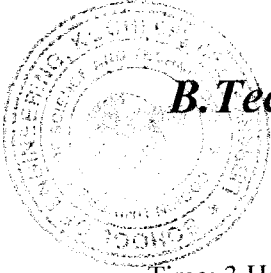


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B.Tech. Degree V Semester Examination November 2018

ME 15-1506 POWER PLANT ENGINEERING (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What are the factors to be considered for selecting location of a thermal power plant?
- (b) What do you mean by load factor? What is the effect of load factor on cost of energy?
- (c) State the essential elements of a hydro electric power plant.
- (d) What is surge tank? State its importance in a hydro power plant.
- (e) Explain the working of an open cycle gas turbine with thermodynamic cycle sketch.
- (f) What are the advantages and disadvantages of a gas turbine plant?
- (g) What are the factors to be considered for selecting the location of a wind turbine plant?
- (h) Differentiate between heterogeneous nuclear reactors and homogeneous nuclear reactors.
- (i) What are the methods employed in the disposal of nuclear wastes?
- (j) Explain the working principle of a tidal energy power plant with basic elements involved.

PART B

(4 × 10 = 40)

- II. (a) The peak load on a power plant is 60 MW. The loads having maximum demands of 30 MW, 20 MW, 10 MW and 14 MW are connected to power plant. The capacity of power plant is 80 MW and the annual load factor is 0.5. Estimate (i) the average load on power plant (ii) the energy supplied per year (iii) the demand factor (iv) diversity factor. (8)
- (b) How can you reduce the cost of power generation? (2)

OR

- III. (a) Explain what you understand by base load and peaking load. Why are base plants loaded heavily? (3)
- (b) What are the advantages of using draft tubes in a hydro electric plant? (3)
- (c) Classify the hydro turbines according to head, power, size and specific speed. (4)

(P.T.O.)

- IV. (a) Explain the methods of improving thermal efficiency of an open cycle gas turbine power plant. (6)
- (b) Explain the effects of turbine inlet temperature and pressure ratio on thermal efficiency of an open cycle gas turbine. (4)

OR

- V. A gas turbine unit receives air at 1 bar and 300 K and compresses it adiabatically to 6.2 bar. The compressor efficiency is 88%. The fuel has a heating value of 44186 KJ/kg and the fuel air ratio is 0.017 KJ/kg of air. The turbine internal efficiency is 90%. Calculate the work of turbine and compressor per kg of air compressed and the thermal efficiency. For products of combustion, take $C_p = 1.147$ KJ/kg K, $\gamma = 1.33$. (10)

- VI. (a) Differentiate between fire tube boilers and water tube boilers. (5)
- (b) What are the advantages of artificial draught over natural draught? (2)
- (c) What are the functions of economizer and super heater in a thermal plant? (3)

OR

- VII. (a) With sketch explain the working of a pressurized water reactor power plant. (6)
- (b) Explain the working of a boiling water reactor (BWR) with internal circulation. (4)

- VIII. (a) Explain the working of a flat plate solar collector with sketch. (5)
- (b) Explain magneto hydro dynamic power generation. (5)

OR

- IX. (a) Distinguish between thermo electric conversion and thermionic conversion. (6)
- (b) Explain the principle of fuel cells. (2)
- (c) How can the energy from biomass be utilized? (2)
